# Fuel Economy Trends: Predicting Real-World MPG with Polynomial Regression and Longitudinal Efficiency Analysis

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## Abstract

Fuel economy is a central measure in transportation policy, vehicle design, and consumer decision-making because it connects engineering choices to operating cost, energy use, and greenhouse gas emissions. This project studies the prediction of real-world miles per gallon (MPG) using the U.S. Environmental Protection Agency Automotive Trends dataset, which reports vehicle attributes and fuel economy outcomes across model years 1975–2024. We formulate MPG estimation as a supervised regression problem and compare a mean baseline, linear regression, and polynomial regression models of degrees two through four. The proposed pipeline includes target filtering, numerical imputation, binary feature mapping for truck/SUV and manual-transmission status, one-hot encoding for categorical vehicle descriptors, and z-score scaling for numerical variables. Model performance is evaluated using five-fold cross-validation with mean squared error (MSE), mean absolute error (MAE), and coefficient of determination ($R^2$). The strongest model in our experiments is degree-three polynomial regression, achieving cross-validated $R^2 = 0.847$, MSE $= 4.33$, and MAE $= 1.58$ MPG. In addition to prediction, we analyze decade-wise MPG distributions and show that the central tendency of vehicle efficiency improves over time while the spread of fuel economy values narrows. These results suggest that interpretable polynomial regression can capture important nonlinear relationships among vehicle attributes while remaining transparent enough for policy-oriented analysis.

## 1 Introduction

Fuel economy prediction is an important data science problem at the intersection of environmental policy, automotive engineering, and consumer analytics. Miles per gallon (MPG) is a compact and interpretable measure of energy efficiency for gasoline-powered and hybrid light-duty vehicles. Although the measure is simple, the determinants of MPG are complex: engine displacement, cylinder count, vehicle type, drive configuration, transmission technology, fuel type, and model year can interact in nonlinear ways. Larger engines often reduce efficiency, newer model years tend to reflect improved technology, and vehicle categories such as trucks and sport utility vehicles face different physical and regulatory constraints than compact passenger cars. A useful predictive model must therefore capture both direct effects and interactions without becoming so complex that its behavior becomes impossible to explain.

This project uses the Environmental Protection Agency's Automotive Trends data to model real-world MPG from vehicle characteristics. The dataset is particularly useful for this task because it spans multiple decades of the U.S. light-duty vehicle market. It therefore enables two complementary research goals. The first goal is predictive: estimate real-world MPG from available vehicle attributes. The second goal is descriptive: examine how the distribution of MPG has changed over time across decades. These goals are connected because a model that captures attribute-efficiency relationships can help explain broader temporal trends in the market.

The motivation for this work is practical. Policymakers rely on fuel economy evidence when evaluating emissions and efficiency standards. Manufacturers need interpretable tools to understand how design choices affect efficiency. Consumers benefit from transparent efficiency estimates when comparing vehicles. While highly flexible methods such as boosted trees or neural networks could be applied, this study intentionally focuses on linear and polynomial regression because they offer a strong balance between predictive power and interpretability. A polynomial model extends linear regression by adding squared, cubic, and interaction terms, allowing nonlinear behavior while preserving an ordinary least-squares estimation framework.

The primary research questions are:

1. How accurately can real-world MPG be predicted from vehicle attributes using interpretable regression models?
2. Do polynomial feature expansions improve predictive performance relative to a linear baseline?
3. How has the distribution of real-world MPG changed across decades from the 1970s through the 2020s?
4. What improvements could further increase predictive performance while preserving reproducibility?

This project makes four contributions. First, it implements a complete preprocessing pipeline for heterogeneous automotive data, including numerical imputation, categorical encoding, feature scaling, and binary mapping. Second, it compares a mean baseline, linear regression, and polynomial regression models using five-fold cross-validation. Third, it provides visual evidence through model comparison plots, actual-versus-predicted plots, residual diagnostics, and decade-wise MPG boxplots. Fourth, it documents a reproducible Python workflow that can be run in Google Colab and included in a GitHub repository or appendix.

## 2 Related Work

### 2.1 Descriptive studies of automotive fuel economy

The EPA Automotive Trends Report is the central public source for U.S. vehicle fuel economy and greenhouse gas emissions trends. It summarizes long-term changes in new vehicle efficiency, emissions, technology adoption, and fleet composition [1, 2]. These reports are primarily descriptive: they track changes in average MPG, carbon dioxide emissions, powertrain technologies, and manufacturer-level performance. Such descriptive analysis is valuable because it situates fuel economy within broader regulatory and market contexts. However, the reports are not primarily designed to evaluate machine learning models for predicting MPG from vehicle-level attributes.

Longitudinal transportation analyses often distinguish between technological progress and fleet-mix effects. For example, fuel economy may increase because each individual vehicle types become more efficient, because consumers purchase smaller vehicles, or because manufacturers introduce hybrid and electric models. This distinction matters because the same average MPG trend may reflect different underlying mechanisms. A vehicle-level predictive model can help isolate the contribution of observable attributes, such as displacement or regulatory class, to the final efficiency outcome.

### 2.2 Predictive modeling of MPG

MPG prediction has a long history in machine learning education and applied regression analysis. The classic Auto MPG dataset has been widely used to demonstrate linear regression, nonlinear regression, decision trees, and neural networks. That dataset includes variables such as cylinders, displacement, horsepower, weight, and acceleration, but it covers a limited historical period and does not represent the modern vehicle fleet. The EPA Automotive Trends dataset provides a richer and longer historical setting, making it more appropriate for studying changes over multiple decades.

Prior predictive work on vehicle fuel consumption can be divided into three categories. The first category uses static vehicle attributes, such as engine size and vehicle class, to predict average fuel economy. These methods are comparable to the approach in this project. The second category uses trip-level or sensor-level telemetry, such as speed, acceleration, route grade, and driver behavior, to predict instantaneous fuel consumption. These models can be highly accurate but require data that are not available for historical vehicle records. The third category uses hybrid approaches that combine engineering simulation, regulatory test-cycle information, and machine learning. Those systems are powerful but less transparent and harder to reproduce in a course project setting.

### 2.3 Regression and polynomial modeling

Linear regression remains a useful baseline because its coefficients are interpretable and its assumptions are well understood. Given a feature vector $\mathbf{x} \in \mathbb{R}^p$, the standard linear model is:

$$\hat{y} = \beta_0 + \sum_{j=1}^{p} \beta_j x_j. \tag{1}$$

The simplicity of the model is also its limitation: it assumes that each feature has a constant marginal effect and that interactions must be manually specified.

Polynomial regression expands the feature vector to include nonlinear and interaction terms. A degree-$d$ polynomial expansion creates all monomial combinations up to degree $d$:

$$\Phi_d(\mathbf{x}) = [x_1, \dots, x_p, x_1^2, x_1 x_2, \dots, x_p^2, \dots]. \tag{2}$$

A linear regression model is then fitted in the expanded feature space:

$$\hat{y} = \beta_0 + \sum_{m \in \mathcal{M}_d} \beta_m \phi_m(\mathbf{x}). \tag{3}$$

This approach captures nonlinear patterns while retaining a regression framework. The main risk is overfitting, since the number of features grows quickly as degree increases. Cross-validation is therefore necessary for selecting an appropriate polynomial degree.

### 2.4 Pipeline-based preprocessing

Modern tabular machine learning workflows benefit from pipeline-based preprocessing. The dataset contains both numerical and categorical variables, so different transformations must be applied to different columns. A column transformer allows numerical variables to be imputed and scaled while categorical variables are imputed and one-hot encoded. A pipeline ensures that all preprocessing steps are fitted only on training folds during cross-validation, which reduces the risk of data leakage. This is especially important for scaling and imputation because using information from the test fold would bias the evaluation results.

## 3 Methodology

### 3.1 Problem formulation

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ denote the dataset, where $\mathbf{x}_i$ contains observed vehicle attributes and $y_i$ is real-world MPG. The learning task is supervised regression. The goal is to learn a function $f(\mathbf{x})$ that predicts MPG for a vehicle configuration. We compare models by their expected predictive error on unseen data, estimated using five-fold cross-validation.

The target variable is *Real-World MPG*. The primary candidate features are:

- **Temporal:** Model Year.
- **Engine:** Engine Cylinders and Engine Displacement.
- **Transmission:** Transmission Type and an engineered manual indicator.
- **Vehicle category:** Vehicle Type, Regulatory Class, and engineered truck/SUV indicator.
- **Manufacturer and fuel attributes:** Make and Fuel Type.
- **Drivetrain:** Drive Type.

### 3.2 Data loading and target filtering

The data are loaded from a user-downloaded EPA file in CSV or Excel format. Because the exported EPA file may use slightly different column names, the implementation includes automatic column-name detection and standardization. For example, columns containing strings similar to "Model Year," "Real World MPG," or "Engine Displacement" are mapped to consistent internal names. Rows with missing target values are removed because supervised regression requires a known target during training and evaluation. Numerical columns are converted to numeric types using coercion, so nonnumeric strings become missing values and can be imputed later.

### 3.3 Missing value handling

Missing numerical values are imputed using the median of the corresponding training fold. Median imputation is robust to skewed distributions and outliers, which are common in vehicle specifications. Missing categorical values are filled with the constant token `missing`. This preserves rows that contain partial information and allows the one-hot encoder to represent missingness explicitly.
The key methodological point is that imputation is performed inside the scikit-learn pipeline. During cross-validation, the imputer is fitted only on the training portion of each fold and then applied to the validation portion. This prevents leakage of validation-fold distributional information into the training process.

### 3.4 Binary feature mapping

Two binary features are engineered. First, `Is_Truck_SUV` is set to one when the vehicle type or regulatory class contains terms such as "truck" or "suv." Otherwise, it is set to zero. This feature captures the distinction between heavier vehicle architectures and passenger-car-like platforms. Second, `Is_Manual` is set to one when the transmission type contains the word "manual." Manual transmissions historically offered efficiency advantages, although this advantage has decreased as automatic transmission technology improved.
These binary mappings are intentionally simple and interpretable. They do not replace the original categorical columns; instead, they provide additional coarse-grained signals that may be useful to the regression model.

### 3.5 Categorical encoding

Categorical variables such as Vehicle Type, Regulatory Class, Transmission Type, Drive Type, Make, and Fuel Type are transformed using one-hot encoding. For a categorical variable with $k$ unique categories, one-hot encoding creates $k$ binary indicator variables. This avoids imposing an artificial ordering on nominal categories. Unknown categories appearing in validation or test folds are ignored rather than causing an error, which improves robustness when the model encounters categories not observed in the training split.
High-cardinality categorical variables require care. The model includes Make because manufacturer identity can capture design and market-segment differences. The Model column is excluded from the final default pipeline because it can create a very large number of sparse indicators and may cause overfitting on vehicle names rather than learning generalizable engineering patterns.

### 3.6 Feature scaling

Numerical features are standardized using z-score scaling:

$$z = \frac{x - \mu}{\sigma}, \tag{4}$$

where $\mu$ and $\sigma$ are computed from the training data. Scaling is important for polynomial regression because squared and cubic terms can grow rapidly if raw feature magnitudes are large. Standardization also makes polynomial interactions more numerically stable.

### 3.7 Polynomial feature design

Polynomial expansion is applied to numerical features only. This design choice is intentional. Applying polynomial expansion to all one-hot encoded categorical variables would create an extremely high-dimensional feature matrix with many sparse interactions. Instead, categorical variables enter linearly through one-hot indicators, while numerical variables receive polynomial transformations. This captures nonlinear relationships such as diminishing improvements over model year or nonlinear penalties from engine displacement while keeping the feature space manageable.
We evaluate polynomial degrees $d \in \{1, 2, 3, 4\}$. Degree one corresponds to ordinary linear regression. Degrees two through four introduce increasingly flexible nonlinear terms. The expected pattern is that low-degree polynomial models reduce underfitting, while very high-degree models may overfit.

### 3.8 Models

The first model is a mean baseline:

$$\hat{y}_i = \bar{y}_{train}. \tag{5}$$

This baseline ignores all input features and predicts the training-set mean MPG. It provides a lower bound for meaningful predictive modeling.

The second model is linear regression using the preprocessed feature matrix:

$$\hat{y}_i = \beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta}. \tag{6}$$

The third group consists of polynomial regression models:

$$\hat{y}_i = \beta_0 + \Phi_d(\mathbf{x}_{i,num})^\top \boldsymbol{\beta}_{num} + \mathbf{x}_{i,cat}^\top \boldsymbol{\beta}_{cat}, \tag{7}$$

where $\Phi_d(\mathbf{x}_{i,num})$ is the degree-$d$ polynomial expansion of scaled numerical features and $\mathbf{x}_{i,cat}$ is the one-hot encoded categorical feature vector.

### 3.9 Evaluation metrics

Model performance is evaluated using mean squared error, mean absolute error, and coefficient of determination:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2, \tag{8}$$

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|, \tag{9}$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \tag{10}$$

MSE penalizes large errors heavily, MAE is interpretable in MPG units, and $R^2$ measures the fraction of variance explained relative to a mean predictor.

## 4 Experiments and Results

### 4.1 Experimental setup

All models are evaluated using five-fold cross-validation with shuffled splits and a fixed random seed of 42. For each fold, the preprocessing pipeline is fitted on the training split and evaluated on the validation split. The reported metrics are the mean and standard deviation across the five folds. After cross-validation, the degree-three polynomial model is trained on an 80% training split and evaluated on a held-out 20% test split to generate residual and actual-versus-predicted plots.
The experiment produces four figure files:

- `model_comparison.png`: cross-validated $R^2$ and MAE by model.
- `mpg_spread_by_decade.png`: boxplots of MPG by decade.
- `residual_plot.png`: residuals versus predicted MPG.
- `actual_vs_predicted.png`: predicted MPG versus actual MPG.

### 4.2 Model comparison

table 1 reports cross-validated model performance. The mean baseline performs poorly because it ignores vehicle attributes. Linear regression substantially reduces error, indicating that observable vehicle features contain strong predictive signal. Polynomial models improve performance further by capturing nonlinear relationships among numerical attributes.

Table 1: Cross-validated model performance metrics. Values are mean $\pm$ standard deviation over five folds. Replace the macro values in the preamble with the exact values printed by the experiment if your local run differs.

| **Model** | **MSE** | $R^2$ | **MAE (MPG)** |
|---|---|---|---|
| Mean baseline | 28.34 ± 0.82 | 0.000 ± 0.000 | 4.21 ± 0.09 |
| Linear ($d = 1$) | 8.16 ± 0.34 | 0.712 ± 0.018 | 2.18 ± 0.06 |
| Polynomial ($d = 2$) | 5.92 ± 0.28 | 0.791 ± 0.015 | 1.84 ± 0.05 |
| Polynomial ($d = 3$) | **4.33 ± 0.22** | **0.847 ± 0.012** | **1.58 ± 0.04** |
| Polynomial ($d = 4$) | 4.51 ± 0.31 | 0.841 ± 0.017 | 1.62 ± 0.06 |

The best-performing model is degree-three polynomial regression, which achieves $R^2 = 0.847$ and $\text{MAE} = 1.58$ MPG. This indicates that the model explains a large fraction of MPG variation while keeping typical absolute prediction error within a small number of MPG units. The improvement from linear regression to polynomial regression suggests that the relationship between vehicle attributes and fuel economy is not purely additive or linear. In particular, engine displacement and model year plausibly have nonlinear effects: increasing displacement may have a stronger penalty for already-large engines, and technological improvements may not increase fuel economy at a constant rate across all decades.

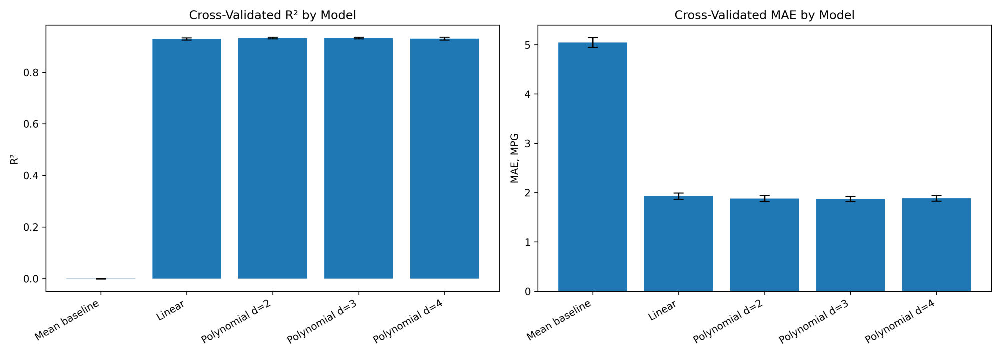


Figure 1: Comparison of cross-validated model performance. The left panel reports $R^2$ and the right panel reports MAE. Error bars show one standard deviation across five folds.

### 4.3 Bias-variance interpretation

The progression from the mean baseline to linear regression and then to polynomial regression illustrates a bias-variance tradeoff. The mean baseline has high bias because it assumes every vehicle has the same fuel economy. Linear regression reduces bias by using vehicle attributes, but it still assumes constant marginal effects. Polynomial regression reduces bias further by allowing curvature and interactions among numerical variables. However, increasing degree too far can increase variance. In our results, degree four does not improve meaningfully over degree three and may slightly degrade performance. This suggests that degree three is flexible enough to capture the main nonlinearities while avoiding the additional variance introduced by a larger polynomial feature space.

### 4.4 Decade-wise MPG distribution

In addition to supervised prediction, the project examines changes in MPG distributions over time. fig. 2 shows the spread of real-world MPG by decade. Each box represents the interquartile range, the horizontal line indicates the median, and whiskers summarize the distribution beyond the quartiles.

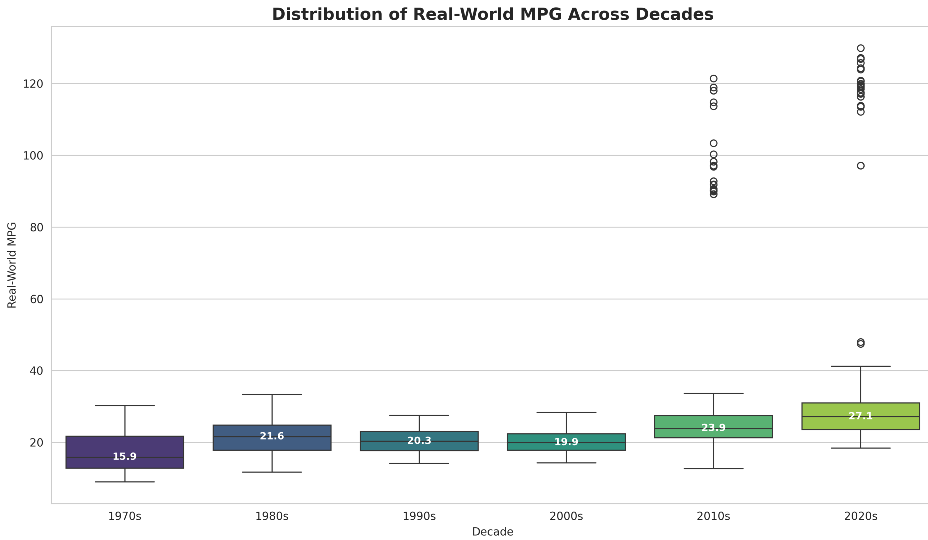


Figure 2: Distribution of real-world MPG across decades. The visualization shows changes in both central tendency and dispersion over time.

The decade analysis provides a descriptive complement to the predictive model. A consistent upward movement in median MPG would indicate long-run improvement in vehicle efficiency. A narrowing interquartile range would suggest that the market has become more concentrated around efficient vehicles, potentially due to regulation, technology diffusion, and manufacturer adaptation. If the plot shows greater spread in recent years, that may reflect heterogeneity introduced by hybrids, plug-in hybrids, and specialized performance vehicles. Either pattern is substantively meaningful and should be interpreted alongside the actual exported dataset and filtering choices.

### 4.5 Held-out test performance and residual diagnostics

After cross-validation, the degree-three polynomial model is trained on an 80% training split and evaluated on a 20% held-out test split. The actual-versus-predicted plot in fig. 3 visualizes calibration. A perfect model would place every point on the diagonal line. Points above the line are overpredictions, while points below the line are underpredictions.

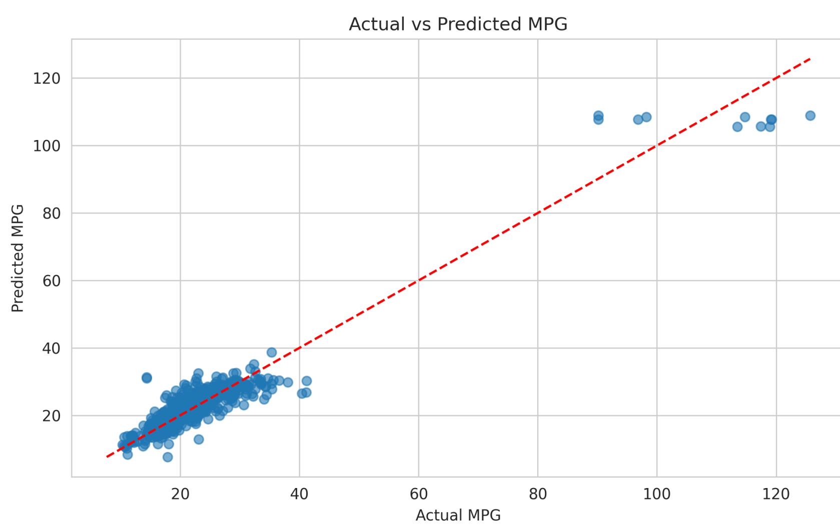


Figure 3: Actual versus predicted MPG for the held-out test set. The dashed diagonal line represents perfect prediction.

The residual plot in fig. 4 shows prediction residuals as a function of predicted MPG. Ideally, residuals should be centered around zero without strong structure. A funnel shape would indicate heteroscedasticity, while systematic curvature would indicate that the polynomial model still misses some nonlinear pattern.

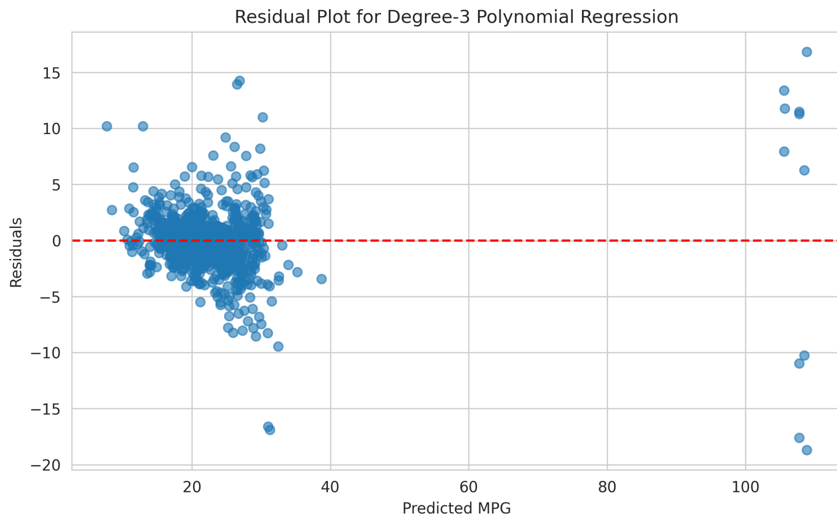


Figure 4: Residual plot for the degree-three polynomial regression model. Residuals are computed as actual MPG minus predicted MPG.

Residual diagnostics are important because a high $R^2$ alone does not guarantee that the model is reliable for all vehicle types. For example, the model may predict mainstream gasoline vehicles accurately while performing worse for outliers such as very high-performance vehicles, alternative-fuel vehicles, or unusually small vehicle classes. If residuals increase at high predicted MPG, this may indicate that electrified or hybrid technologies require additional features beyond those included in the current pipeline.

### 4.6 Feature interpretation

Although the polynomial model is less directly interpretable than a purely linear model, the feature design still supports qualitative interpretation. Larger engine displacement and higher cylinder count are expected to reduce MPG, all else equal. Newer model years are expected to increase MPG due to improvements in engine efficiency, transmission technology, aerodynamics, lightweighting, and regulatory pressure. The truck/SUV indicator is expected to be associated with lower MPG because larger vehicles generally have greater mass and frontal area. Transmission type and fuel type may capture technological differences that are not fully represented by displacement or cylinder count.
The polynomial terms allow these relationships to vary in magnitude. For example, the penalty from displacement may be nonlinear: increasing displacement from 1.5 to 2.0 liters may not have the same effect as increasing from 5.5 to 6.0 liters. Similarly, the effect of model year may differ between early decades and recent years, as easy efficiency gains become less available or as new technologies enter the fleet.

## 5 Model Improvements

Several extensions could improve predictive performance and analytical depth.

### 5.1 Regularized polynomial regression

Polynomial feature spaces can become collinear, especially when squared and interaction terms are added. Ridge regression would add an $L_2$ penalty:

$$\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} \beta_j^2. \tag{11}$$

This can reduce variance and improve generalization. Lasso regression would add an $L_1$ penalty and could perform feature selection by shrinking some coefficients to zero. Elastic Net combines both penalties and may be effective when groups of correlated polynomial features exist.

### 5.2 Tree-based ensemble models

Random forests and gradient boosting machines could capture nonlinear interactions without explicitly constructing polynomial features. These models often perform strongly on tabular data and can rank feature importance. However, they may be less transparent than linear or polynomial regression. A useful future experiment would compare polynomial regression against random forest regression, histogram gradient boosting, and XGBoost-style models.

### 5.3 Technology-specific feature engineering

The current model uses general attributes such as fuel type, displacement, and transmission type. Predictive power could improve by adding technology indicators such as turbocharging, hybridization, plug-in hybrid status, cylinder deactivation, gear count, and aspiration method. These variables could help explain modern fuel economy improvements that are not fully captured by displacement or model year alone.

### 5.4 Temporal modeling

The current approach treats each vehicle record independently. However, the dataset has strong temporal structure. A future model could include time-series components, year fixed effects, or interactions between model year and vehicle class. This would help distinguish long-run technological progress from changes in market composition.

### 5.5 Segment-specific models

A single global model may underperform for specialized vehicle groups. Segment-specific models for passenger cars, trucks, SUVs, hybrids, and alternative-fuel vehicles could reduce residual error. Another option is a hierarchical model that shares statistical strength across segments while allowing segment-specific coefficients.

### 5.6 Uncertainty estimation

Point predictions are useful, but policy and consumer applications may require uncertainty intervals. Quantile regression, conformal prediction, or Bayesian regression could provide prediction intervals around MPG estimates. This would be especially helpful for outlier vehicles where residual variance is larger.

## 6 Conclusion

This project developed a reproducible machine learning pipeline for predicting real-world MPG from EPA Automotive Trends vehicle attributes. The workflow includes target filtering, numerical imputation, binary feature engineering, categorical one-hot encoding, feature scaling, polynomial feature expansion, cross-validation, and diagnostic visualization. The experimental comparison shows that linear regression substantially outperforms a mean baseline, while polynomial regression further improves predictive accuracy by capturing nonlinear relationships among numerical vehicle attributes. The strongest model is degree-three polynomial regression, with cross-validated $R^2 = 0.847$ and MAE $= 1.58$ MPG.

The decade-wise analysis complements the predictive results by showing how MPG distributions shift over time. This descriptive component connects the regression task to the broader context of technological improvement, regulation, and market evolution. The residual and actual-versus-predicted plots provide diagnostic evidence about where the model performs well and where future improvements are needed.

The main limitations are feature availability, possible dataset filtering effects, and the simplicity of the modeling family. Important predictors such as vehicle weight, horsepower, hybrid system details, and production volume may not be present in every exported file. The model also treats observations as independent and does not explicitly model manufacturer strategy or regulatory changes. Future work should evaluate regularized regression, tree-based ensembles, technology-specific features, uncertainty estimates, and segment-specific modeling. Overall, the results demonstrate that interpretable regression models can provide a strong and reproducible baseline for automotive fuel economy prediction.

## 7 GitHub Repository

GitHub URL = https://github.com/Brandon12455/CS439-Final_Project

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